

PYTHON ASSIGNMENT 2 – PYTHON

LIST, DICTIONARY, SET & CONDITIONAL STATEMENTS

TOOLS: Python / VS Code

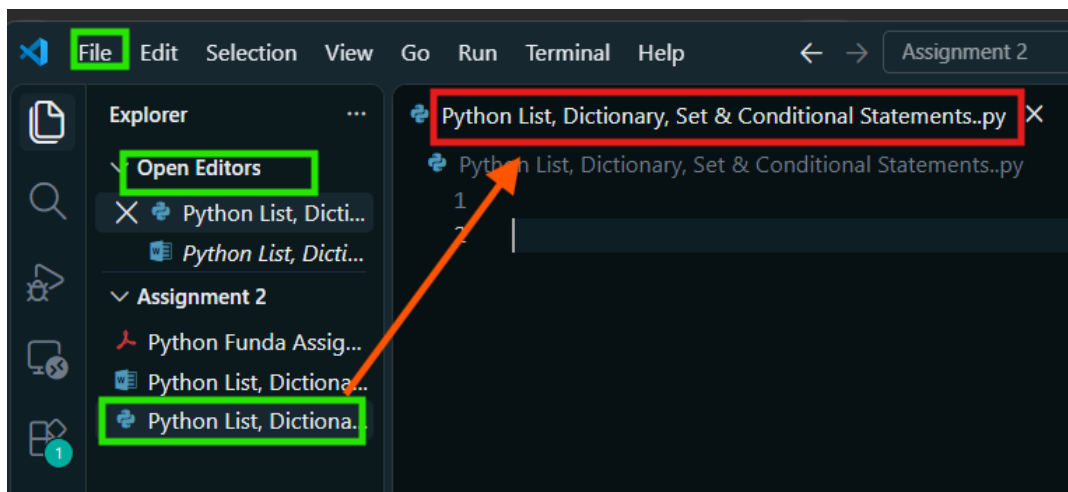
PROJECT TITLE: Python Data Structures – List, Dictionary, Set & Conditional Statements

PROJECT OBJECTIVE: Working with Lists, Dictionaries and Sets – creation, modification and access – and implementing Conditional Statements (if, elif, else)

DATE OF SUBMISSION: 07/10/2026 – 1:30 PM

FILE SETUP:

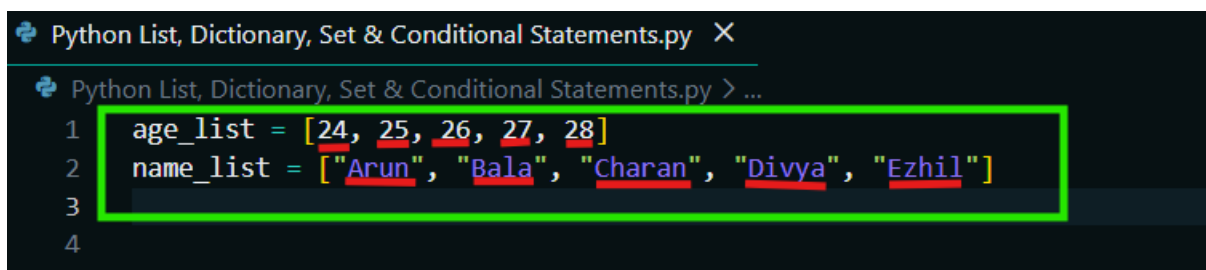
CREATE A NEW PYTHON FILE "PYTHON LIST, DICTIONARY, SET & CONDITIONAL STATEMENTS.PY" IN VS CODE.



LIST (CREATION, MODIFICATION AND ACCESS):

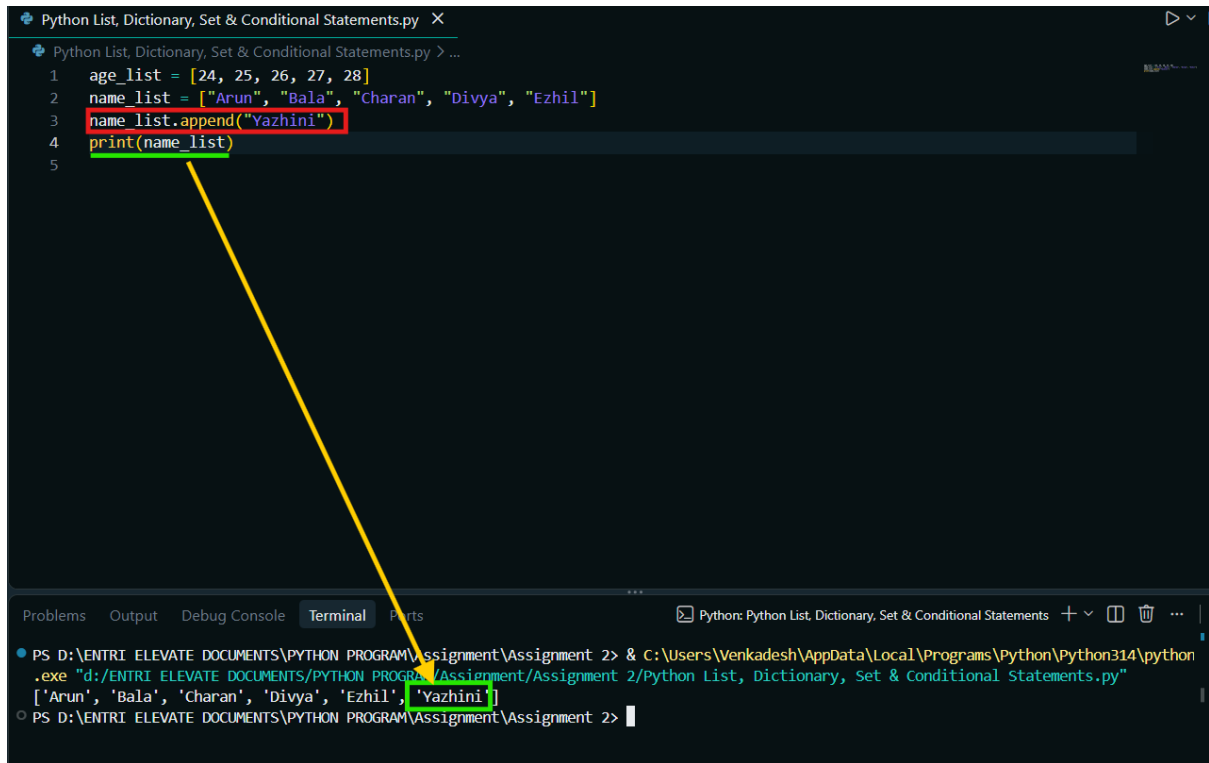
1) LIST CREATION:

1.1) CREATE A LIST "AGE_LIST" WITH FIVE INTEGER ELEMENTS AND A LIST "NAME_LIST" WITH FIVE STRING ELEMENTS.



2) LIST OPERATIONS / MODIFICATIONS:

2.1) APPEND THE STRING "YAZHINI" TO "NAME_LIST" USING "APPEND()".



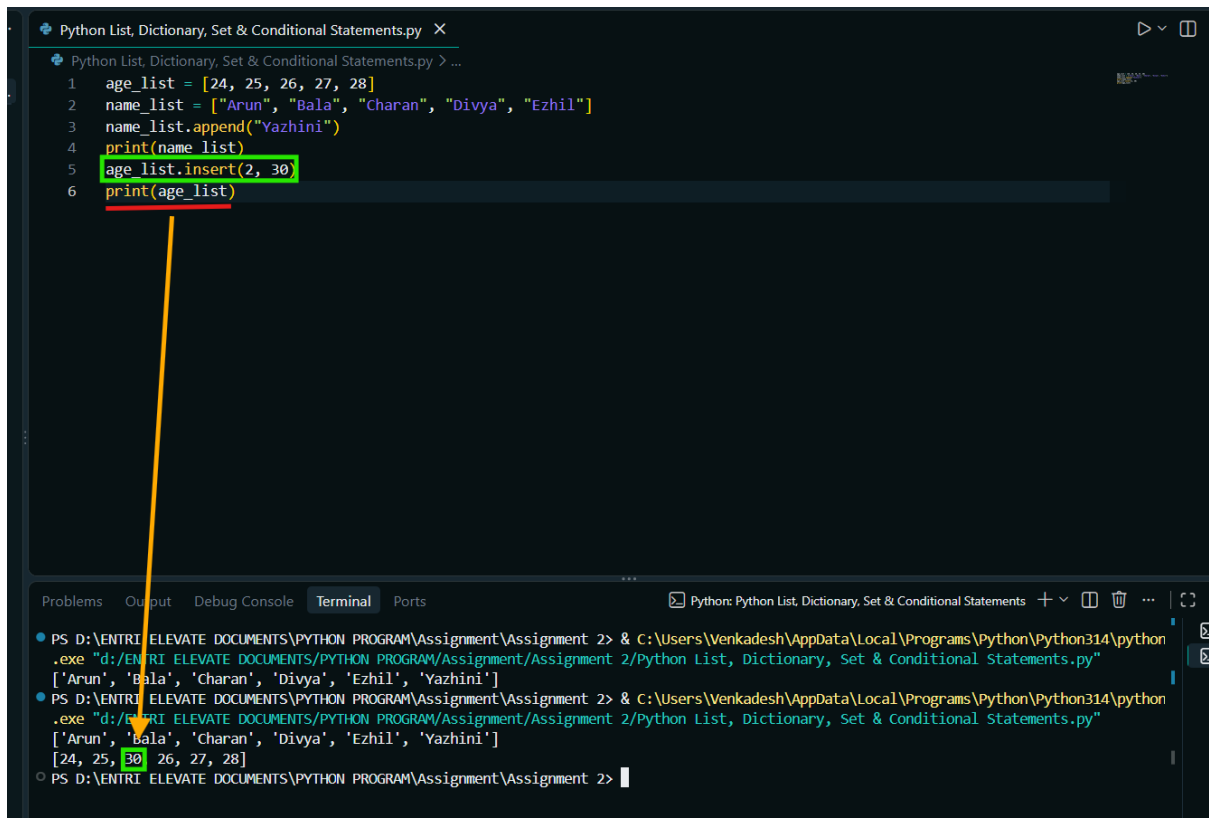
The screenshot shows a Python IDE with a file named "Python List, Dictionary, Set & Conditional Statements.py". The code is as follows:

```
1 age_list = [24, 25, 26, 27, 28]
2 name_list = ["Arun", "Bala", "Charan", "Divya", "Ezhil"]
3 name_list.append("Yazhini")
4 print(name_list)
5
```

A yellow arrow points from the `print(name_list)` statement in the code to the terminal output. The terminal shows the execution of the script, resulting in the following output:

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statements.py"
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil', 'Yazhini']
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

2.2) INSERT THE ELEMENT 30 AT INDEX 2 IN "AGE_LIST" USING "INSERT()".



The screenshot shows the same Python IDE with the same file. The code is now:

```
1 age_list = [24, 25, 26, 27, 28]
2 name_list = ["Arun", "Bala", "Charan", "Divya", "Ezhil"]
3 name_list.append("Yazhini")
4 print(name_list)
5 age_list.insert(2, 30)
6 print(age_list)
```

A yellow arrow points from the `print(age_list)` statement in the code to the terminal output. The terminal shows the execution of the script, resulting in the following output:

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statements.py"
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil', 'Yazhini']
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statements.py"
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil', 'Yazhini']
[24, 25, 30, 26, 27, 28]
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

2.3) REMOVE THE STRING "YAZHINI" FROM "NAME_LIST" USING "REMOVE()" AND POP THE LAST ELEMENT FROM "AGE_LIST" USING "POP()".

```
Python List, Dictionary, Set & Conditional Statements.py X
Python List, Dictionary, Set & Conditional Statements.py > ...
1 age_list = [24, 25, 26, 27, 28]
2 name_list = ["Arun", "Bala", "Charan", "Divya", "Ezhil"]
3 name_list.append("Yazhini")
4 print(name_list)
5 age_list.insert(2, 30)
6 print(age_list)
7 name_list.remove("Yazhini")
8 print(name_list)
9 age_list.pop()
10 print(age_list)
11
```

Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & Conditional Statements

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Programs\Python\Py
.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statemen
[24, 25, 30, 26, 27, 28]
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil']
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Programs\Python\Py
.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statemen
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil', 'Yazhini']
[24, 25, 30, 26, 27, 28]
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil']
[24, 25, 30, 26, 27]
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

2.4) EXTEND "AGE_LIST" WITH [29, 30, 26] USING "EXTEND()" AND SORT IT IN DESCENDING ORDER USING "SORT(REVERSE=TRUE)".

```
Python List, Dictionary, Set & Conditional Statements.py > ...
1  age_list = [24, 25, 26, 27, 28]
2  name_list = ["Arun", "Bala", "Charan", "Divya", "Ezhil"]
3  name_list.append("Yazhini")
4  print(name_list)
5  age_list.insert(2, 30)
6  print(age_list)
7  name_list.remove("Yazhini")
8  print(name_list)
9  age_list.pop()
10 print(age_list)
11 age_list.extend([29, 30, 26])
12 print(age_list)
13 age_list.sort(reverse=True)
14 print(age_list)
15
```

Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & Cond

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Pro
.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Condi
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Pro
.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Condi
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil', 'Yazhini']
[24, 25, 30, 26, 27, 28]
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil']
[24, 25, 30, 26, 27]
[24, 25, 30, 26, 27, 29, 30, 26]
[30, 30, 29, 27, 26, 26, 25, 24]
```

2.5) FIND THE MAXIMUM AGE, MINIMUM AGE AND SUM OF ALL AGES USING "MAX()", "MIN()" AND "SUM()".

```
Python List, Dictionary, Set & Conditional Statements.py > ...
1  age_list = [24, 25, 26, 27, 28]
2  name_list = ["Arun", "Bala", "Charan", "Divya", "Ezhil"]
3  name_list.append("Yazhini")
4  print(name_list)
5  age_list.insert(2, 30)
6  print(age_list)
7  name_list.remove("Yazhini")
8  print(name_list)
9  age_list.pop()
10 print(age_list)
11 age_list.extend([29, 30, 26])
12 print(age_list)
13 age_list.sort(reverse=True)
14 print(age_list)
15 print("Age list:", age_list)
16 print("Maximum age:", max(age_list))
17 print("Minimum age:", min(age_list))
18 print("Sum of ages:", sum(age_list))
19
```

Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & C

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\
.exe "d:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2\python_list_dictionary_set_2.py"
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil']
[24, 25, 30, 26, 27]
[24, 25, 30, 26, 27, 29, 30, 26]
[30, 30, 29, 27, 26, 26, 25, 24]
Age list: [30, 30, 29, 27, 26, 26, 25, 24]
Maximum age: 30
Minimum age: 24
Sum of ages: 217
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

Navigate to Command
Scroll to Previous Command (Ctrl+UpArrow)
Scroll to Next Command (Ctrl+DownArrow)

(SUMMARY) COMPLETE CODE AND OUTPUT OF ALL THE LIST MODIFICATIONS.

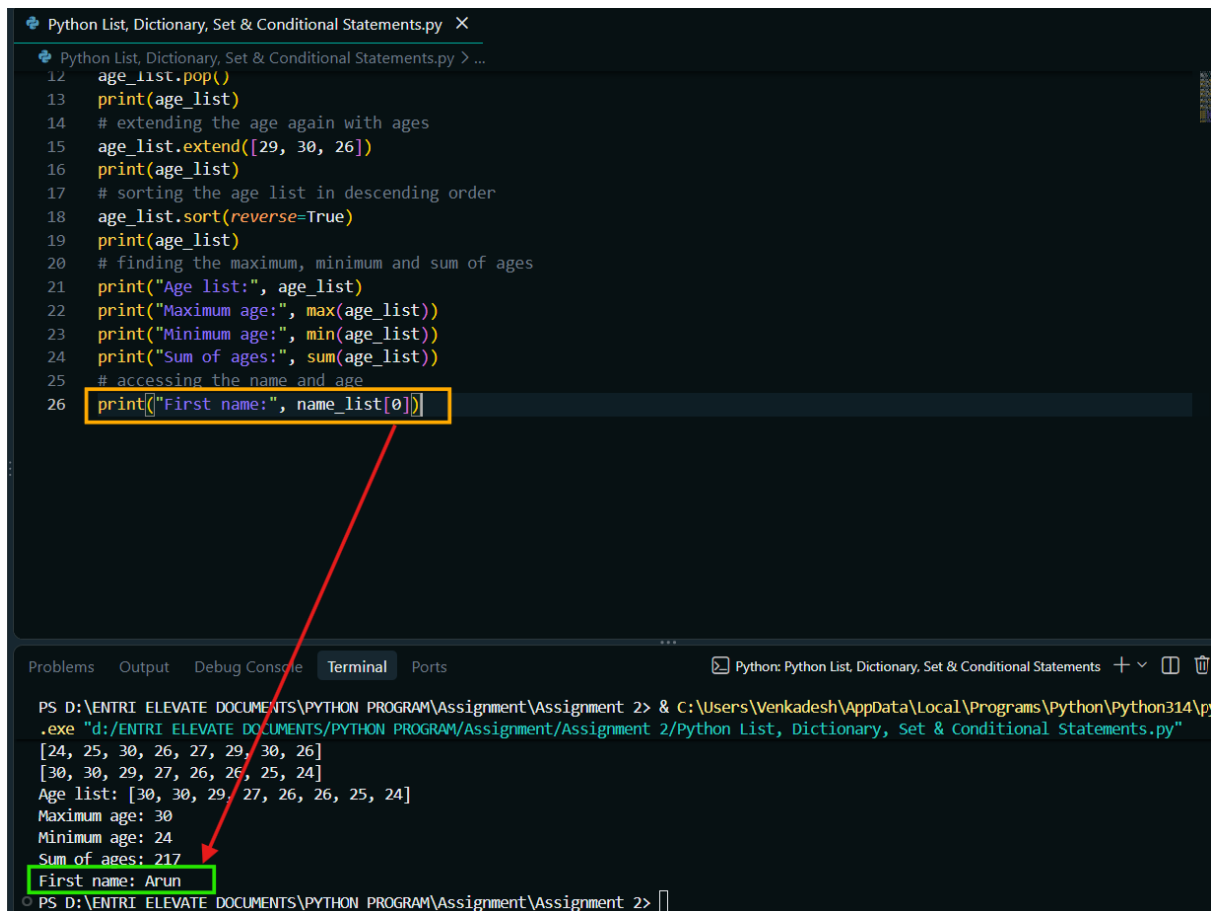
```
Python List, Dictionary, Set & Conditional Statements.py X
Python List, Dictionary, Set & Conditional Statements.py > ...
1 # List of Name and Age
2 age_list = [24, 25, 26, 27, 28]
3 name_list = ["Arun", "Bala", "Charan", "Divya", "Ezhil"]
4 # adding a new name and age to the list
5 name_list.append("Yazhini")
6 print(name_list)
7 age_list.insert(2, 30)
8 print(age_list)
9 # removing a name and age from the list
10 name_list.remove("Yazhini")
11 print(name_list)
12 age_list.pop()
13 print(age_list)
14 # extending the age again with ages
15 age_list.extend([29, 30, 26])
16 print(age_list)
17 # sorting the age list in descending order
18 age_list.sort(reverse=True)
19 print(age_list)
20 # finding the maximum, minimum and sum of ages
21 print("Age list:", age_list)
22 print("Maximum age:", max(age_list))
23 print("Minimum age:", min(age_list))
24 print("Sum of ages:", sum(age_list))
```

Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & Conditional Statements + - [] [X]

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe "d:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2\Python List, Dictionary, Set & Conditional Statements.py"
[24, 25, 30, 26, 27]
[24, 25, 30, 26, 27, 29, 30, 26]
[30, 30, 29, 27, 26, 26, 25, 24]
Age list: [30, 30, 29, 27, 26, 26, 25, 24]
Maximum age: 30
Minimum age: 24
Sum of ages: 217
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

3) ACCESSING LIST ELEMENTS:

3.1) PRINT THE FIRST ELEMENT OF "NAME_LIST" USING INDEX "[0]".



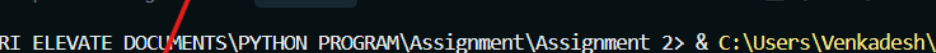
```
Python List, Dictionary, Set & Conditional Statements.py X
Python List, Dictionary, Set & Conditional Statements.py > ...
12 age_list.pop()
13 print(age_list)
14 # extending the age again with ages
15 age_list.extend([29, 30, 26])
16 print(age_list)
17 # sorting the age list in descending order
18 age_list.sort(reverse=True)
19 print(age_list)
20 # finding the maximum, minimum and sum of ages
21 print("Age list:", age_list)
22 print("Maximum age:", max(age_list))
23 print("Minimum age:", min(age_list))
24 print("Sum of ages:", sum(age_list))
25 # accessing the name and age
26 print("First name:", name_list[0])
```

Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & Conditional Statements + v

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statements.py"
[24, 25, 30, 26, 27, 29, 30, 26]
[30, 30, 29, 27, 26, 26, 25, 24]
Age list: [30, 30, 29, 27, 26, 26, 25, 24]
Maximum age: 30
Minimum age: 24
Sum of ages: 217
First name: Arun
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

3.2) PRINT THE LAST ELEMENT OF "NAME_LIST" USING NEGATIVE INDEX "[-1]".

```
Python List, Dictionary, Set & Conditional Statements.py X
Python List, Dictionary, Set & Conditional Statements.py > ...
12 age_list.pop()
13 print(age_list)
14 # extending the age again with ages
15 age_list.extend([29, 30, 26])
16 print(age_list)
17 # sorting the age list in descending order
18 age_list.sort(reverse=True)
19 print(age_list)
20 # finding the maximum, minimum and sum of ages
21 print("Age list:", age_list)
22 print("Maximum age:", max(age_list))
23 print("Minimum age:", min(age_list))
24 print("Sum of ages:", sum(age_list))
25 # accessing the name and age
26 # accessing the first name using positive indexing
27 print("First name:", name_list[0])
28 # accessing the last name using negative indexing
29 print("Last name:", name_list[-1])
```



```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Microsoft\Windows\Common-IntelliSense\Python\Python.exe "d:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2\Python List, Dictionary, Set & Tuple.py"
Age list: [30, 30, 29, 27, 26, 26, 25, 24]
Maximum age: 30
Minimum age: 24
Sum of ages: 217
First name: Arun
Last name: Ezhil
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```


3.3) PRINT THE ELEMENTS FROM INDEX 2 TO INDEX 4 IN "NAME_LIST" USING THE SLICE "[2:5]".

```
Python List, Dictionary, Set & Conditional Statements.py X
Python List, Dictionary, Set & Conditional Statements.py > ...
12 age_list.pop()
13 print(age_list)
14 # extending the age again with ages
15 age_list.extend([29, 30, 26])
16 print(age_list)
17 # sorting the age list in descending order
18 age_list.sort(reverse=True)
19 print(age_list)
20 # finding the maximum, minimum and sum of ages
21 print("Age list:", age_list)
22 print("Maximum age:", max(age_list))
23 print("Minimum age:", min(age_list))
24 print("Sum of ages:", sum(age_list))
25 # accessing the name and age
26 # accessing the first name using positive indexing
27 print("First name:", name_list[0])
28 # accessing the last name using negative indexing
29 print("Last name:", name_list[-1])
30 # accessing the names from index 2 to 4 using slicing
31 print("Names at indexes 2 to 4:", name_list[2:5])
```

Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & Conditional Statements.py

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.10.0.0_q-bz882sh6jv1q\python.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statements.py"
Age list: [30, 30, 29, 27, 26, 26, 25, 24]
Maximum age: 30
Minimum age: 24
Sum of ages: 217
First name: Arun
Last name: Ezhil
Names at indexes 2 to 4: ['Charan', 'Divya', 'Ezhil']
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

3.4) PRINT THE ELEMENTS OF "NAME_LIST" IN REVERSE ORDER USING THE SLICE "[::-1]"

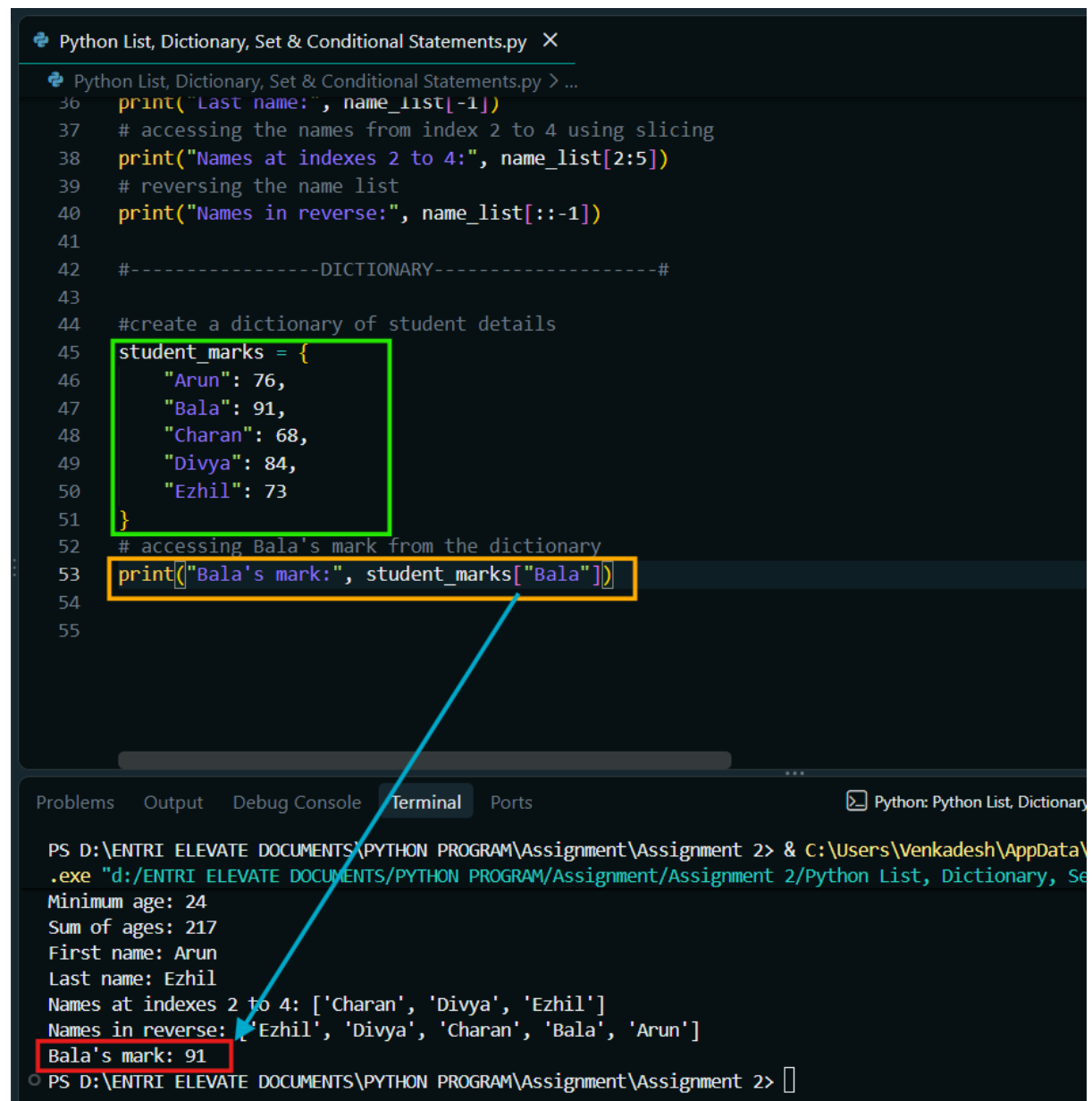
```
Python List, Dictionary, Set & Conditional Statements.py X
Python List, Dictionary, Set & Conditional Statements.py > ...
25 print("Maximum age:", max(age_list))
26 print("Minimum age:", min(age_list))
27 print("Sum of ages:", sum(age_list))
28
29 # 3. accessing the name and age
30 # accessing the first name using positive indexing
31 print("First name:", name_list[0])
32 # accessing the last name using negative indexing
33 print("Last name:", name_list[-1])
34 # accessing the names from index 2 to 4 using slicing
35 print("Names at indexes 2 to 4:", name_list[2:5])
36 # reversing the name list
37 print("Names in reverse:", name_list[::-1])
```

Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & Conditional Statements + - [] ...

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe "d:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2\Python List, Dictionary, Set & Conditional Statements.py"
Maximum age: 30
Minimum age: 24
Sum of ages: 217
First name: Arun
Last name: Ezhil
Names at indexes 2 to 4: ['Charan', 'Divya', 'Ezhi']
Names in reverse: ['Ezhi', 'Divya', 'Charan', 'Bala', 'Arun']
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

4) DICTIONARY (CREATION, MODIFICATION AND ACCESS):

4.1) CREATE A DICTIONARY "**STUDENT_MARKS**" MAPPING FIVE STUDENTS TO THEIR MARKS AND PRINT THE MARK OF "**BALA**".



```
Python List, Dictionary, Set & Conditional Statements.py X
Python List, Dictionary, Set & Conditional Statements.py > ...
36 print("Last name:", name_list[-1])
37 # accessing the names from index 2 to 4 using slicing
38 print("Names at indexes 2 to 4:", name_list[2:5])
39 # reversing the name list
40 print("Names in reverse:", name_list[::-1])
41
42 #-----DICTIONARY-----#
43
44 #create a dictionary of student details
45 student_marks = {
46     "Arun": 76,
47     "Bala": 91,
48     "Charan": 68,
49     "Divya": 84,
50     "Ezhil": 73
51 }
52 # accessing Bala's mark from the dictionary
53 print("Bala's mark:", student_marks["Bala"])
54
55
```

Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & Conditional Statements.py

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.9.0.0_qbkcsq0w3oqpvm9q3w27hkr615h7\python.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statements.py"
Minimum age: 24
Sum of ages: 217
First name: Arun
Last name: Ezhil
Names at indexes 2 to 4: ['Charan', 'Divya', 'Ezhil']
Names in reverse: ['Ezhil', 'Divya', 'Charan', 'Bala', 'Arun']
Bala's mark: 91
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

4.2) ADD A NEW STUDENT "JANANI" WITH MARK 80, UPDATE THE MARK OF "CHARAN" TO 82 AND PRINT ALL KEYS, VALUES AND ITEMS USING "KEYS()", "VALUES()" AND "ITEMS()".

```
Python List, Dictionary, Set & Conditional Statements.py X
Python List, Dictionary, Set & Conditional Statements.py > ...

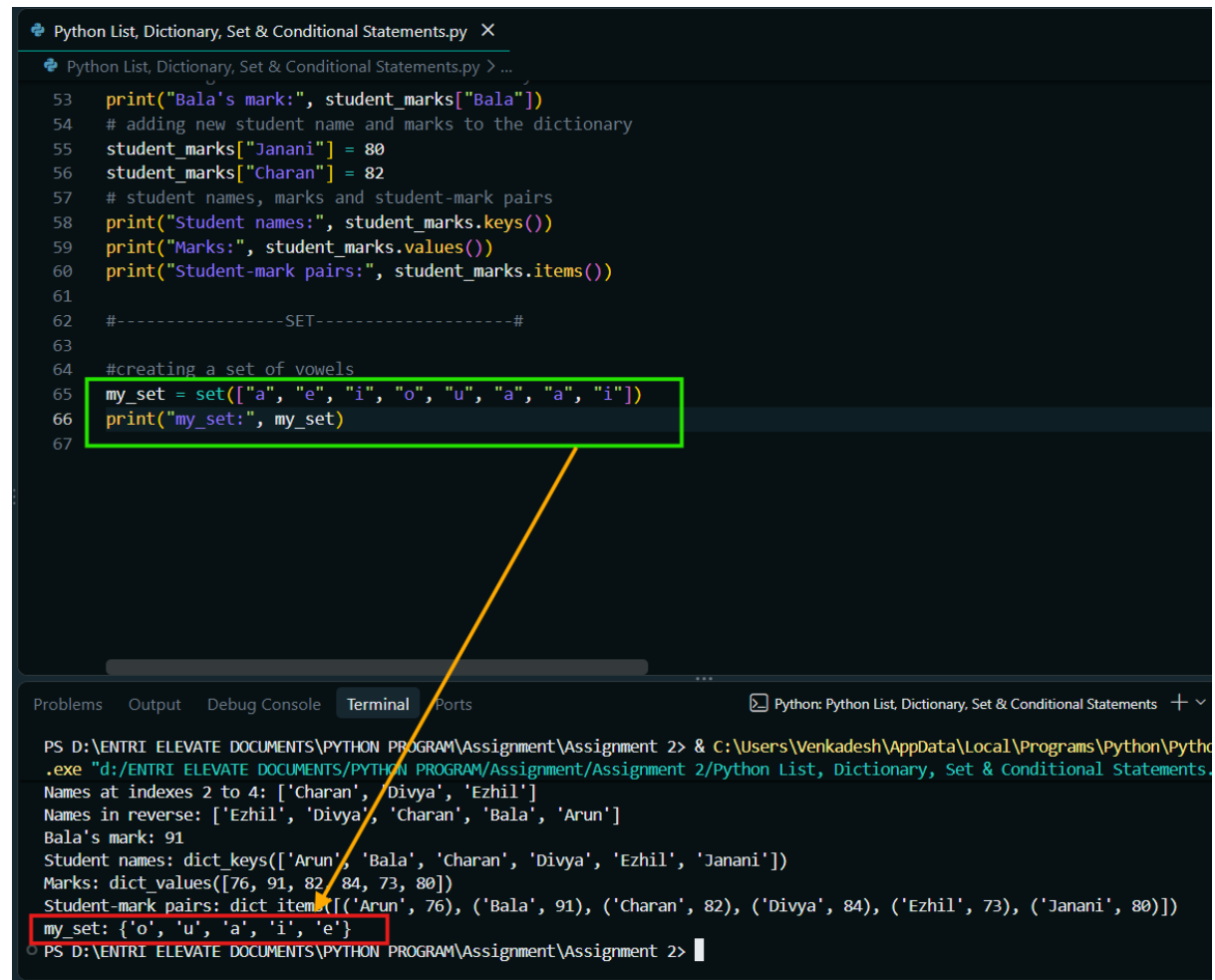
42 #-----DICTIONARY-----#
43
44 #create a dictionary of student details
45 student_marks = {
46     "Arun": 76,
47     "Bala": 91,
48     "Charan": 68,
49     "Divya": 84,
50     "Ezhil": 73
51 }
52 # accessing Bala's mark from the dictionary
53 print("Bala's mark:", student_marks["Bala"])
54 # adding new student name and marks to the dictionary
55 student_marks["Janani"] = 80
56 student_marks["Charan"] = 82
57 # student names, marks and student-mark pairs
58 print("Student names:", student_marks.keys())
59 print("Marks:", student_marks.values())
60 print("Student-mark pairs:", student_marks.items())
61
```

Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & Conditional Statements + v

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe "d:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2\Python List, Dictionary, Set & Conditional Statements.py"
Last name: Ezhil
Names at indexes 2 to 4: ['Charan', 'Divya', 'Ezhil']
Names in reverse: ['Ezhil', 'Divya', 'Charan', 'Bala', 'Arun']
Bala's mark: 91
Student names: dict_keys(['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil', 'Janani'])
Marks: dict_values([76, 91, 82, 84, 73, 80])
Student-mark pairs: dict_items([('Arun', 76), ('Bala', 91), ('Charan', 82), ('Divya', 84), ('Ezhil', 73), ('Janani', 80)])
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

5) SETS (OPERATIONS):

5.1) CREATE A SET "MY_SET" FROM THE VALUES A, E, I, O, U, A, A, I AND PRINT IT.



The screenshot shows a Python IDE with a file named "Python List, Dictionary, Set & Conditional Statements.py". The code defines a dictionary of student marks and creates a set of vowels. The terminal output shows the execution results, with the final line highlighting the output of the set creation.

```
53 print("Bala's mark:", student_marks["Bala"])
54 # adding new student name and marks to the dictionary
55 student_marks["Janani"] = 80
56 student_marks["Charan"] = 82
57 # student names, marks and student-mark pairs
58 print("Student names:", student_marks.keys())
59 print("Marks:", student_marks.values())
60 print("Student-mark pairs:", student_marks.items())
61
62 #-----SET-----#
63
64 #creating a set of vowels
65 my_set = set(["a", "e", "i", "o", "u", "a", "a", "i"])
66 print("my_set:", my_set)
67
```

Terminal Output:

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Programs\Python\Python38\python.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statements.py"
Names at indexes 2 to 4: ['Charan', 'Divya', 'Ezhil']
Names in reverse: ['Ezhil', 'Divya', 'Charan', 'Bala', 'Arun']
Bala's mark: 91
Student names: dict_keys(['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil', 'Janani'])
Marks: dict_values([76, 91, 82, 84, 73, 80])
Student-mark pairs: dict_items([('Arun', 76), ('Bala', 91), ('Charan', 82), ('Divya', 84), ('Ezhil', 73), ('Janani', 80)])
my_set: {'o', 'u', 'a', 'i', 'e'}
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```

5.2) TRY TO CHANGE THE VALUE OF "MY_SET[4]" TO 'S' USING A "TRY-EXCEPT" BLOCK.

```
Python List, Dictionary, Set & Conditional Statements.py X
Python List, Dictionary, Set & Conditional Statements.py > ...
54 # adding new student name and marks to the dictionary
55 student_marks["Janani"] = 80
56 student_marks["Charan"] = 82
57 # student names, marks and student-mark pairs
58 print("Student names:", student_marks.keys())
59 print("Marks:", student_marks.values())
60 print("Student-mark pairs:", student_marks.items())
61
62 #-----SET-----#
63
64 #creating a set of vowels
65 my_set = set(["a", "e", "i", "o", "u", "a", "a", "i"])
66 print("my_set:", my_set)
67 # adding a new vowel to the set
68 try:
69     my_set[4] = "s"
70 except TypeError:
71     print("A set cannot be changed using an index.")
72     print("Sets are unordered and do not support item assignment.")
```

Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & Conditional Statements + - []

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\
.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statements.py"
Bala's mark: 91
Student names: dict_keys(['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil', 'Janani'])
Marks: dict_values([76, 91, 82, 84, 73, 80])
Student-mark pairs: dict_items([('Arun', 76), ('Bala', 91), ('Charan', 82), ('Divya', 84), ('Ezhil', 73), ('Janani', 80)])
my_set: {'i', 'e', 'u', 'o', 'a'}
A set cannot be changed using an index.
Sets are unordered and do not support item assignment.
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> |
```

5.3) CREATE "SET1" AND "SET2" AND PRINT THEIR UNION USING THE "|" OPERATOR AND THEIR INTERSECTION USING THE "&" OPERATOR.

```
Python List, Dictionary, Set & Conditional Statements.py X
Python List, Dictionary, Set & Conditional Statements.py > ...
59 print("Marks:", student_marks.values())
60 print("Student-mark pairs:", student_marks.items())
61
62 #-----SET-----#
63
64 # 1. creating a set of vowels
65 my_set = set(["a", "e", "i", "o", "u", "a", "a", "i"])
66 print("my_set:", my_set)
67 # 2. adding a new vowel to the set
68 try:
69     my_set[4] = "s"
70 except TypeError:
71     print("A set cannot be changed using an index.")
72     print("Sets are unordered and do not support item assignment.")
73 # 3. performing union and intersection of two sets
74 set1 = {1, 3, 5, 7, 9}
75 set2 = {2, 3, 5, 8, 10}
76 print("Union:", set1 | set2)
77 print("Intersection:", set1 & set2)
78
```

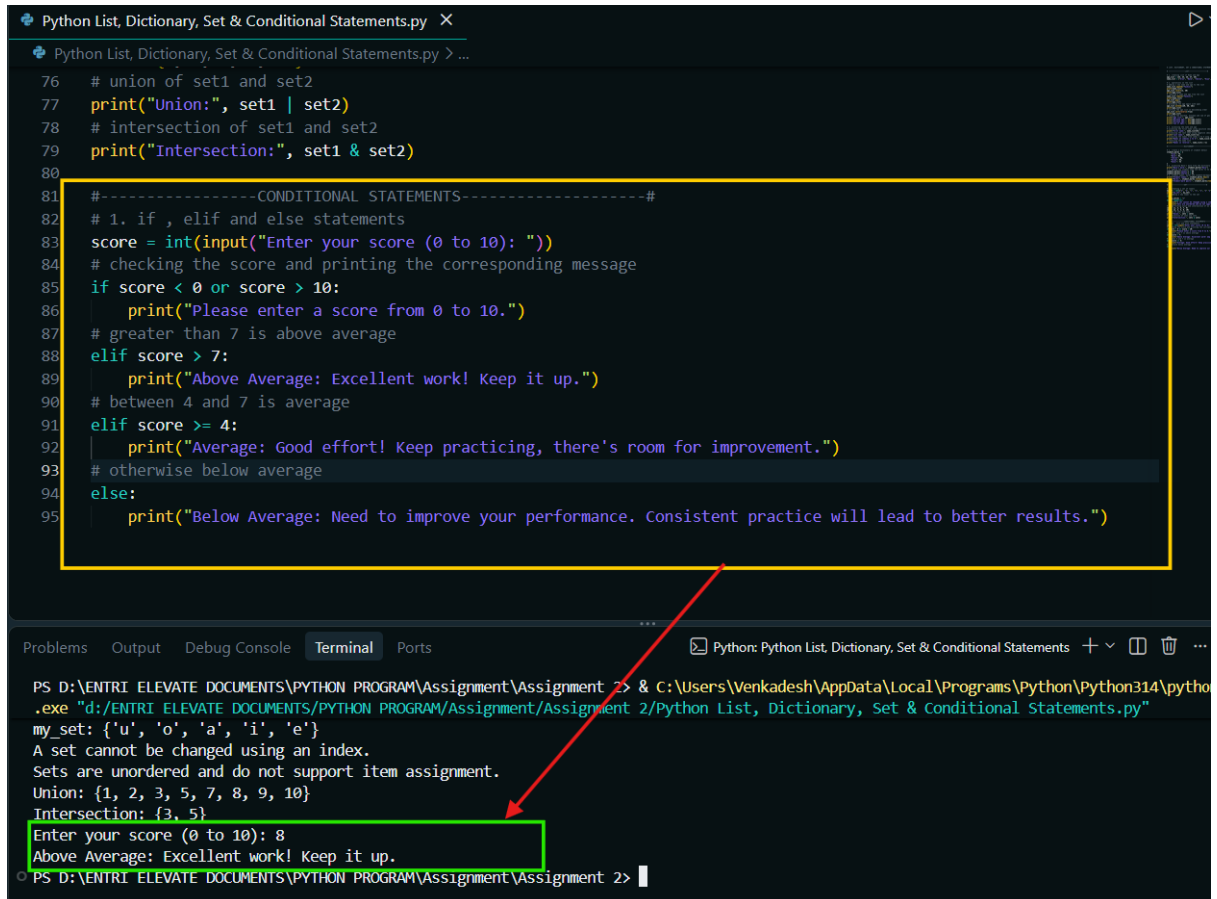
Problems Output Debug Console Terminal Ports Python: Python List, Dictionary, Set & Conditional State

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Programs\Py
.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional S
Marks: dict_values([76, 91, 82, 84, 73, 80])
Student-mark pairs: dict_items([('Arun', 76), ('Bala', 91), ('Charan', 82), ('Divya', 84), ('Ezhil', 73), ('Janani',
my_set: {'a', 'e', 'i', 'u', 'o'})
A set cannot be changed using an index.
Sets are unordered and do not support item assignment.
Union: {1, 2, 3, 5, 7, 8, 9, 10}
Intersection: {3, 5}
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> |
```

6) OPERATORS & CONDITIONAL STATEMENTS (IF, ELIF, ELSE):

PERFORMANCE CATEGORY PROGRAM:

6.1) READ A SCORE FROM 0 TO 10 AND PRINT THE PERFORMANCE CATEGORY USING "IF", "ELIF" AND "ELSE".



The screenshot displays a Python IDE with a file named 'Python List, Dictionary, Set & Conditional Statements.py'. The code defines two sets, 'set1' and 'set2', and prints their union and intersection. It then implements a conditional statement to categorize a score entered by the user. The score is read as an integer, and the program checks if it is less than 0 or greater than 10. If so, it prompts the user to enter a valid score. If the score is greater than 7, it prints 'Above Average: Excellent work! Keep it up.'. If the score is between 4 and 7, it prints 'Average: Good effort! Keep practicing, there's room for improvement.'. Otherwise, it prints 'Below Average: Need to improve your performance. Consistent practice will lead to better results.'.

```
76 # union of set1 and set2
77 print("Union:", set1 | set2)
78 # intersection of set1 and set2
79 print("Intersection:", set1 & set2)
80
81 #-----CONDITIONAL STATEMENTS-----#
82 # 1. if , elif and else statements
83 score = int(input("Enter your score (0 to 10): "))
84 # checking the score and printing the corresponding message
85 if score < 0 or score > 10:
86     print("Please enter a score from 0 to 10.")
87 # greater than 7 is above average
88 elif score > 7:
89     print("Above Average: Excellent work! Keep it up.")
90 # between 4 and 7 is average
91 elif score >= 4:
92     print("Average: Good effort! Keep practicing, there's room for improvement.")
93 # otherwise below average
94 else:
95     print("Below Average: Need to improve your performance. Consistent practice will lead to better results.")
```

The terminal output shows the execution of the program. It displays the union and intersection of the sets, followed by the prompt 'Enter your score (0 to 10): 8'. The program then prints 'Above Average: Excellent work! Keep it up.'.

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> & C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe "d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python List, Dictionary, Set & Conditional Statements.py"
my_set: {'u', 'o', 'a', 'i', 'e'}
A set cannot be changed using an index.
Sets are unordered and do not support item assignment.
Union: {1, 2, 3, 5, 7, 8, 9, 10}
Intersection: {3, 5}
Enter your score (0 to 10): 8
Above Average: Excellent work! Keep it up.
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2>
```


SOURCE CODE –

```
# LIST, DICTIONARY, SET & CONDITIONAL STATEMENTS

# -----LIST-----#

# 1. creating List of Name and Age
age_list = [24, 25, 26, 27, 28]
name_list = ["Arun", "Bala", "Charan", "Divya", "Ezhil"]
# 2. operations on the list
# adding a new name and age to the list
name_list.append("Yazhini")
print(name_list)
age_list.insert(2, 30)
print(age_list)
# removing a name and age from the list
name_list.remove("Yazhini")
print(name_list)
age_list.pop()
print(age_list)
# extending the age again with ages
age_list.extend([29, 30, 26])
print(age_list)
# sorting the age list in descending order
age_list.sort(reverse=True)
print(age_list)
# finding the maximum, minimum and sum of ages
print("Age list:", age_list)
print("Maximum age:", max(age_list))
print("Minimum age:", min(age_list))
print("Sum of ages:", sum(age_list))
# 3. accessing the name and age
# accessing the first name using positive indexing
print("First name:", name_list[0])
# accessing the last name using negative indexing
print("Last name:", name_list[-1])
# accessing the names from index 2 to 4 using slicing
print("Names at indexes 2 to 4:", name_list[2:5])
# reversing the name list
print("Names in reverse:", name_list[::-1])

#-----DICTIONARY-----#

# 1. create a dictionary of student details
student_marks = {
    "Arun": 76,
    "Bala": 91,
    "Charan": 68,
    "Divya": 84,
```

```

    "Ezhi": 73
}
# 2. accessing Bala's mark from the dictionary
print("Bala's mark:", student_marks["Bala"])
# 3. adding new student name and marks to the dictionary
student_marks["Janani"] = 80
student_marks["Charan"] = 82
# 4. student names, marks and student-mark pairs
print("Student names:", student_marks.keys())
print("Marks:", student_marks.values())
print("Student-mark pairs:", student_marks.items())

#-----SET-----#

# 1. creating a set of vowels
my_set = set(["a", "e", "i", "o", "u", "a", "a", "i"])
print("my_set:", my_set)
# 2. adding a new vowel to the set
try:
    my_set[4] = "s"
except TypeError:
    print("A set cannot be changed using an index.")
    print("Sets are unordered and do not support item assignment.")
# 3. performing union and intersection of two sets
set1 = {1, 3, 5, 7, 9}
set2 = {2, 3, 5, 8, 10}
# union of set1 and set2
print("Union:", set1 | set2)
# intersection of set1 and set2
print("Intersection:", set1 & set2)

#-----CONDITIONAL STATEMENTS-----#
# 1. if , elif and else statements
score = int(input("Enter your score (0 to 10): "))
# checking the score and printing the corresponding message
if score < 0 or score > 10:
    print("Please enter a score from 0 to 10.")
# greater than 7 is above average
elif score > 7:
    print("Above Average: Excellent work! Keep it up.")
# between 4 and 7 is average
elif score >= 4:
    print("Average: Good effort! Keep practicing, there's room for improvement.")
# otherwise below average
else:
    print("Below Average: Need to improve your performance. Consistent practice will lead to better results.")

```

OUTPUT –

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 2> &
C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe
"d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 2/Python
List, Dictionary, Set & Conditional Statements.py"
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil', 'Yazhini']
[24, 25, 30, 26, 27, 28]
['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil']
[24, 25, 30, 26, 27]
[24, 25, 30, 26, 27, 29, 30, 26]
[30, 30, 29, 27, 26, 26, 25, 24]
Age list: [30, 30, 29, 27, 26, 26, 25, 24]
Maximum age: 30
Minimum age: 24
Sum of ages: 217
First name: Arun
Last name: Ezhil
Names at indexes 2 to 4: ['Charan', 'Divya', 'Ezhil']
Names in reverse: ['Ezhil', 'Divya', 'Charan', 'Bala', 'Arun']
Bala's mark: 91
Student names: dict_keys(['Arun', 'Bala', 'Charan', 'Divya', 'Ezhil',
'Janani'])
Marks: dict_values([76, 91, 82, 84, 73, 80])
Student-mark pairs: dict_items([('Arun', 76), ('Bala', 91), ('Charan', 82),
('Divya', 84), ('Ezhil', 73), ('Janani', 80)])
my_set: {'e', 'a', 'o', 'i', 'u'}
A set cannot be changed using an index.
Sets are unordered and do not support item assignment.
Union: {1, 2, 3, 5, 7, 8, 9, 10}
Intersection: {3, 5}
Enter your score (0 to 10): 9
Above Average: Excellent work! Keep it up.
```